Theta-Quench Schwinger Lab

A Manuscript-Style Summary of Dynamics, Magic, and Learnability

Abstract

We implement exact small-system theta-quench dynamics for the open-boundary lattice Schwinger mo
evaluate whether stabilizer complexity predicts neural learnability degradation. The pipeline computes
Loschmidt diagnostics, entanglement, and stabilizer Renyi magic, then fits time snapshots with NNQS r
under controlled metrics (validation NLL, KL, TV). Cross-family stress tests extending to TFIM, XXZ, and
ANNNI support a corridor-scoped entropy-controlled magic-to-learnability slope law, while strict all-regi
universality is explicitly falsified and boundary-mapped.

Metadata

- Build date: 2026-03-02
- Primary outputs root: outputs
- Claim style: scoped empirical mechanism with explicit failure-region mapping
- Audience: reviewer-facing technical summary (not a theorem-level universal claim)

1. Introduction and Scope

This report summarizes an end-to-end computational study connecting nonequilibrium gauge dynamics t
complexity diagnostics and machine-learning learnability. The central scientific question is whether post-
quench non-stabilizerness (quantified by stabilizer Renyi magic) predicts optimization hardness in neural
quantum state fitting.

The analysis is deliberately claim-gated: each proposed claim is tied to explicit numerical checks,
uncertainty estimates, and boundary mapping where the relation fails. This avoids universal overstateme
while preserving mechanistic signal where evidence is strong.

2. Methods and Statistical Protocol

Physics model: staggered-fermion lattice Schwinger Hamiltonian after Gauss-law elimination with open boundaries, evolved under sudden theta quenches from ED-prepared ground states. Time evolution uses dense/Krylov paths with cross-checks.

Measured diagnostics: Loschmidt rate function, staggered mass proxy, connected correlators, half-chain Neumann entropy, and stabilizer Renyi magic M2. Learnability is measured via NNQS snapshot fitting (validation NLL primary; KL/TV secondary).

Statistical protocol: bootstrap confidence intervals, permutation tests with FDR correction, mixed/covaria controls (including entropy), and corridor filtering by dynamical intensity. Output claims are accepted onl when gate conditions pass.

3. Numerical Validation Snapshot

Hermiticity check H(theta0)/H(theta1): True/True.

Norm drift max_t ||psi(t)||^2-1 = 3.109e-15; dense-vs-Krylov mismatch = 9.667e-15.

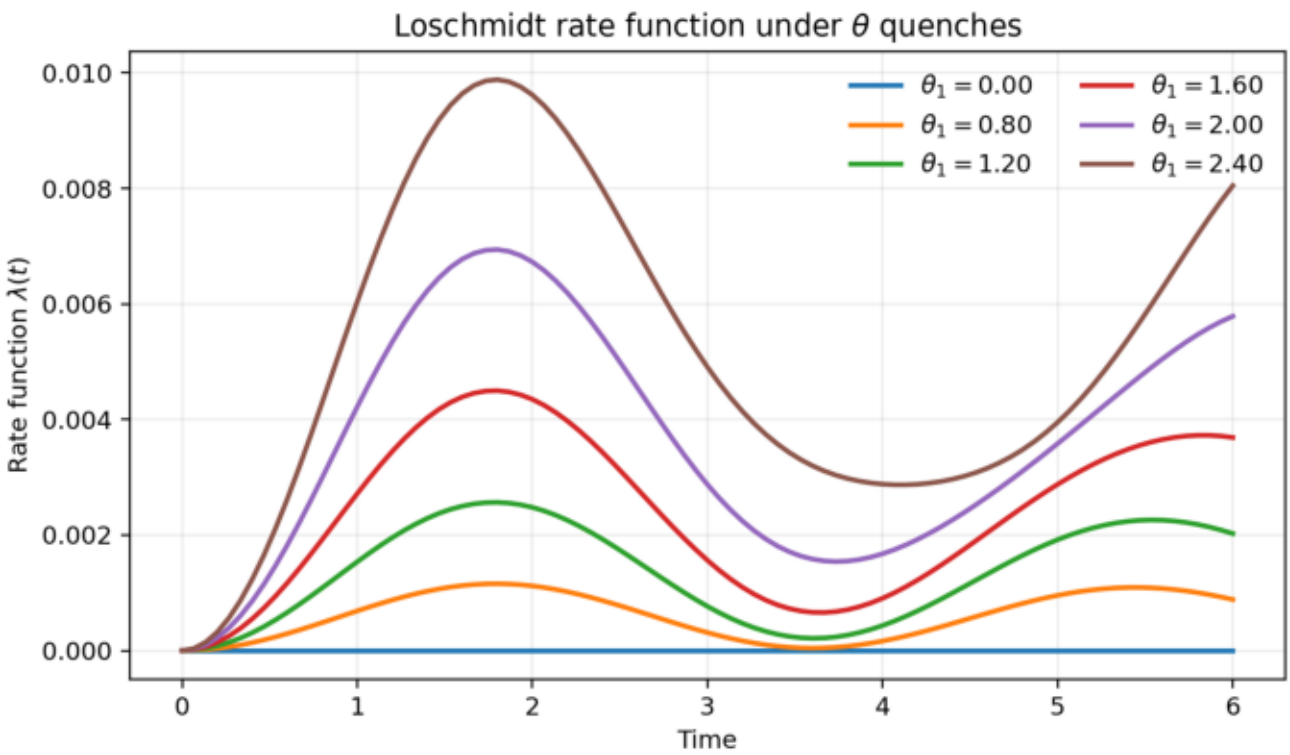
Quench monotonic trends: corr(theta1, max lambda) = 0.9569, corr(theta1, max M2) = 0.9678. NNQS tre
corr(M2, final val NLL) = 0.8202.

4. Claim-Gate Summary Table

Gate	Status	Evidence File	Interpretation
Validation (Hermiticity + unitarit	PASS	outputs/validation.js	Core simulator numerics are co
Stage 3 cross-family beta law	PASS	outputs/stage3_prx/	Cell-level slope positivity in co
Stage 4 thermo + quantile gates	PASS	outputs/stage4_prl/t	Finite-size extrapolation and ha
Stage 5 external family cell gate	PASS	outputs/stage5_prx/	Cross-family mechanism survi
Stage 5 strict regime minimax	FAIL	outputs/stage5_prx/	Explicit failure regions rejects glob

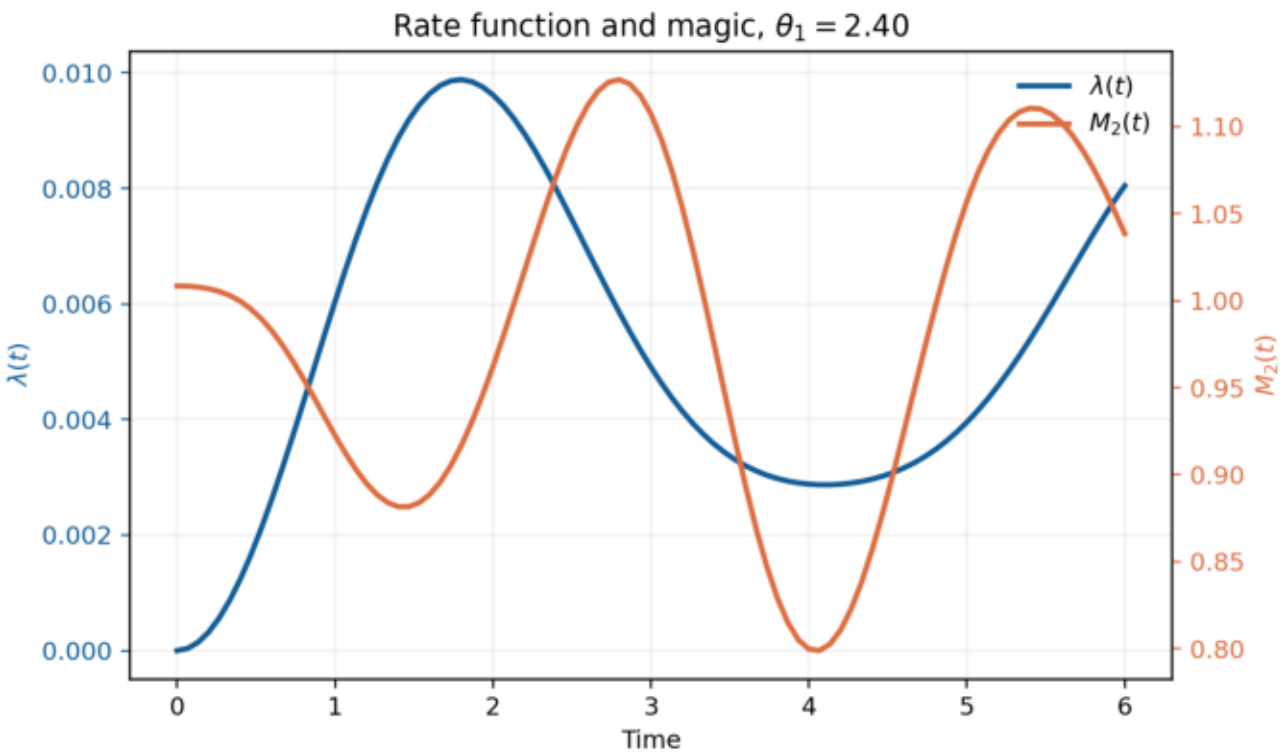
Table 1. Gate-level decision summary. The mechanism is robust in the declared corridor and across model famil
while strict all-regime universality is rejected by explicit minimax/regime failures.

Figure 1. Quench Transition Diagnostic



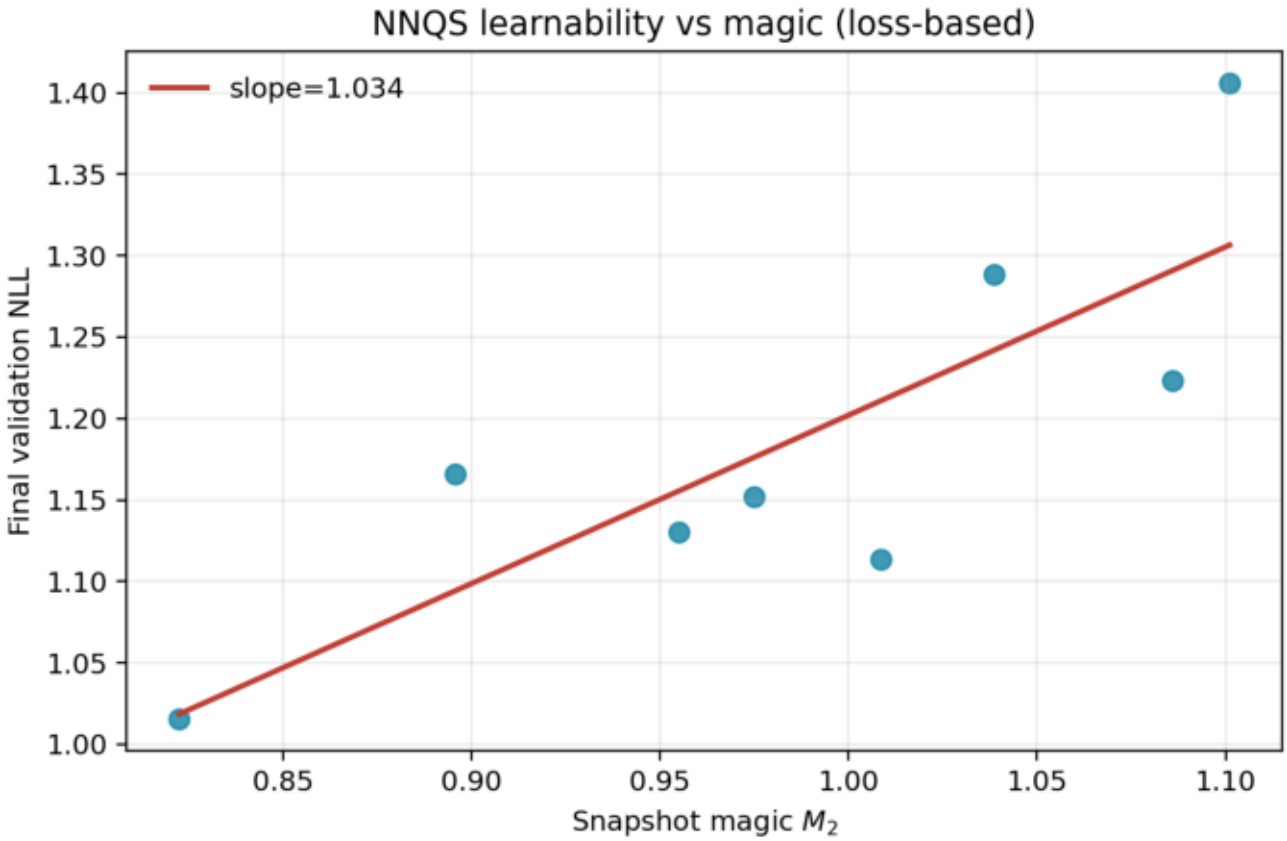
Loschmidt rate family over quench endpoint. Transition-like sharpening appears for stronger quenches.

Figure 2. Complexity Dynamics



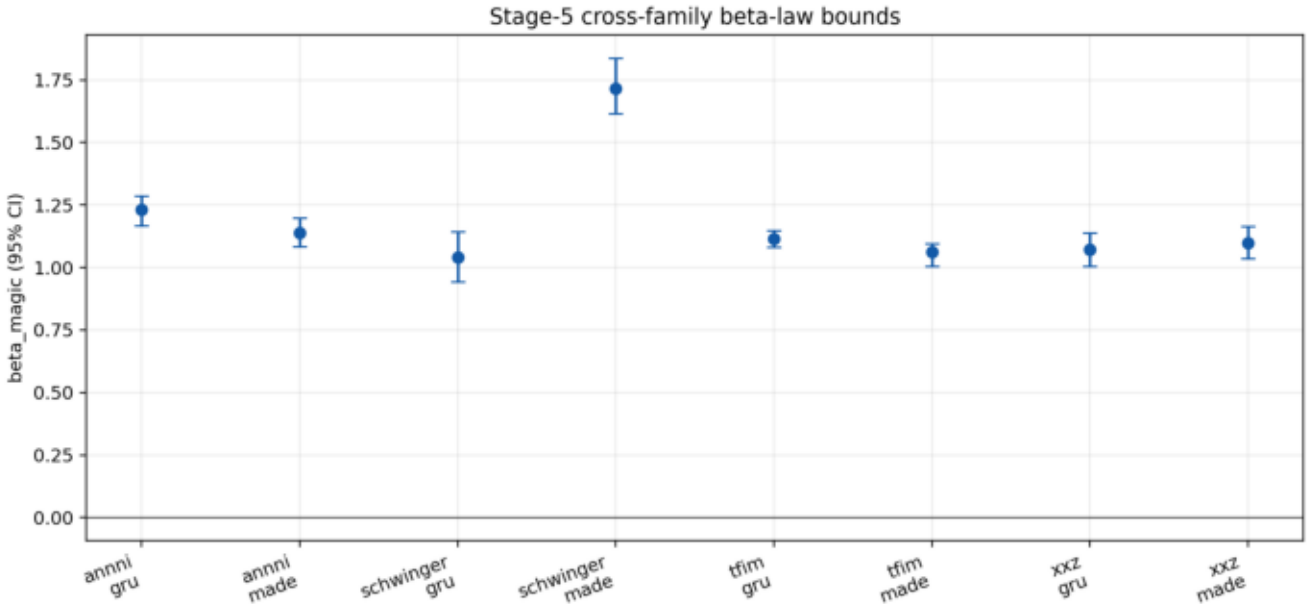
Time-resolved overlay of Loschmidt rate and stabilizer magic M2(t).

Figure 3. Learnability Coupling



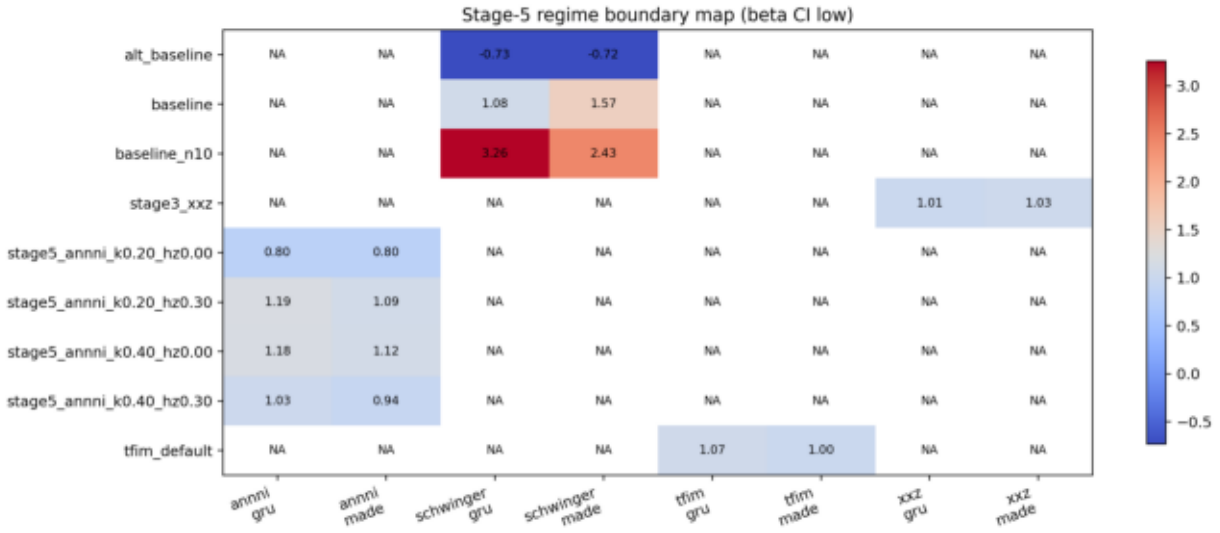
Primary endpoint visualization: higher snapshot magic tends to align with higher final validation NLL.

Figure 4. Cross-Family Cell-Level Bounds



Cell-level entropy-controlled beta bounds after external-family expansion (ANNNI included).

Figure 5. Regime Boundary Map



Regime-level lower CI map for beta_magic, explicitly marking support vs failure regions.

5. Discussion and Publication Position

Final supported claim: in a declared dynamical corridor, higher stabilizer magic predicts worse NNQS learnability after entropy control, robust at cross-family cell level. This is publishable as a scoped empirical law with mechanism-oriented evidence.

Not supported: a model-independent theorem-level universal law, and strict all-regime positivity. The failure region is not hidden; it is explicitly mapped and reported.

This report is therefore intended for scientific review under conservative claim discipline: robust where supported, transparent where boundaries are hit.